



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: July 31, 2026

In addition to part-I (General Handout for all courses) printed on page 1 of the Timetable book, this portion gives further specific details regarding the course.

Course No. : BIO F211
Course Title : Biological Chemistry
Instructors : Prof. Meghana Tare (Instructor in-charge)
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1. Course description: This course is intended to offer an introduction to the world of Biological Chemistry. The course deals with the structure, properties, and significance of the major groups of biochemical compounds (amino acids, proteins, carbohydrates, nucleic acids, and lipids), the bioenergetics, enzyme mechanisms and regulation of the central metabolic pathways. Efforts will be made to include the recent advances and methodologies related to biochemistry. This would help students going for higher level activities, appreciation of biochemical problems, evaluation and problem solving.

2. Text Book (TB): Mary K. Campbell and Shawn O. Farrell, Biochemistry, 9th edition, International edition, copyright 2018 (Thompson Brooks/Cole, Cengage Learning)

3. Reference Books (RB): a) D. Nelson & M Cox. Lehninger, Principles of Biochemistry, Palgrave, 4th Ed. b) Donald Voet et al., Biochemistry, Wiley.

Other teaching resource/material may also be used which will be shared by instructors.

4. Attendance

Attendance is required in order to gain understanding to various aspects of the course. Attendance will be monitored based on appearing in classroom. Besides, surprise evaluation components will be held during class/tutorial hours, for which, no make-up will be given.

5. Course Plan:

Week	Module No.	Lecture Sessions (Tentative)	Reference	Learning Outcomes
1	I	L1-3 Chemical nature of Biomolecules, organization of Prokaryotes and Eukaryotes, how cells use energy.	T1	Orientation to the course and Introduction to Biological Chemistry
2	II.	L4-L6 Chemistry of carbohydrates	T16	Understanding the Carbohydrates Chemistry



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		Structure and stereochemistry, reactions, some important oligosaccharides and polysaccharides and their biological roles		
3	III.	L7-L8 Nucleic acid structure, Nucleic acid chemistry, other functions of Nucleic Acids	R(a) 8, T9	Understanding the Biochemistry of Nucleic Acids
4-5	IV	L9-L16 Amino acids and peptides, three dimensional structure of proteins, protein purification and characterization techniques, Enzymes, Enzymes Kinetics and mechanism of action, Inhibitors and regulators, Allosteric enzymes, Isoenzymes	T3,4,5,6	Amino acids, Peptides and Proteins Learn the Enzyme types, apply understanding of protein structure to understand mechanism of catalytic action, regulation and inhibition.
6	V	L17-L19 Structure and Functions of Lipids, Lipids and proteins associated in membranes, membrane structure	T8	Understanding the lipid structure, functions and Biological membranes
7-8	VI	L20-25 Glycolysis, `Reversal of Glycolytic sequence Utilization of carbohydrates, Regulation of glycolysis, Pentose phosphate pathway, Citric acid cycle, Glyoxylic acid cycle	T17,18, 19	Learn (a) role(s) of glucose and other isomers in the cell and apply towards utilization and production of energy; (b) differential glucose metabolism in diseased conditions
9-10	VII	L26-L30 Hydrolysis and transport of fats, β -oxidation, Oxidation of unsaturated fatty acids, Formation of Ketone bodies, Biosynthesis of fatty acids.	T21	Apply the structural aspects of lipid biochemistry in learning the metabolism of lipids and differentiating the metabolism in normal and diseased conditions
11	VIII	L31-33 Purine Biosynthesis, Pyrimidine Biosynthesis, Salvage pathway.	T23	Apply the understanding of structure of nucleic acids to learn the metabolism of nitrogen.
12	IX	L34-L37 Catabolism of amino acids, Assimilation of ammonia, Urea cycle and formation of uric acid	T23	Apply the understanding of structure of amino acids to learn the metabolism of nitrogen; fate of nucleic and amino acids in the body.
13	X	L37-L40 Components involved in ETC, Respiratory chain Bioenergetics and thermodynamics, chemical logic and common biochemical reactions, phosphoryl groups transfer and ATP, biological oxidation-reduction reactions	T20	Understand and learn the Oxidative phosphorylation and apply to different mechanisms to produce energy in the cell. Learn about Bioenergetics and apply it to different types of Biochemical reactions



14	XI	L41-42 Path of Carbon-Calvin cycle, C4 pathway	T22	Learn the Biochemistry involved in Photosynthesis in plants.
15	XII	L43 Summary: Integration of metabolic pathways	T24	Integration of Metabolic Pathways

6. Evaluation Scheme

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid Semester Test	90 minutes	30	<TEST_1>	CB
Comprehensive Examination	3 h	35	<TEST_C>	Partly OB
Quizzes	During tutorial hours/will be announced	25	TBD	CB
Assignments/Class Discussions/In class tests [#]	Surprise	10	TBD	OB or CB

Please note that of evaluation components will be held during class/tutorial hours and may or may not be announced. (Max. Marks=200 for the course)

No Make ups will be given and class attendance is mandatory for evaluation of these components.

7. Grading Policy: Award of grades would be guided by the histogram of marks and course average. NC Criteria: **If a student obtains below 30% of the average of the top three performers or 40% of median, whichever is lower, the student will not be awarded a valid grade and the case will be reported as “NC”.**

8. Chamber Consultation Hour: Fridays 5-6 PM, prior appointment/intimation is required.

9. Notices: Notices regarding announced components will be made available on Nalanda or email. It is responsibility of students to watch Nalanda portal for the same

10. Make-up Policy: If a student misses the Mid-Semester or Comprehensive Examination due to a genuine reason (serious medical causes leading to hospitalization (certified by doctor from BITS medical center) or personal/family emergencies there exists a provision to apply for make-up. Prior permission must be taken from the Instructor in- Charge, before applying for make-up. No Make-Up will be granted for the evaluation components or activities that are held in lecture/tutorial classes. The decision to grant make-up or not is taken by the instructor team/department DCA and shall be final.

Meghana Tare
Instructor-in-charge
Course No. BIO F211